

Problem Solving Tutorial 1 2025

Notes for tutors and model answers

The aim of the problem solving tutorial is for you to work through practise problems with the help of a tutor who can explain the answers, answer your questions and clarify any points where you got stuck. We want these sessions to be as interactive as possible, so please do ask the tutor questions (and share your questions with the group) during the class.

To get the most out of the session, please attempt the questions marked **[Attempt before the PST]** in advance of the PST session. The tutor will work through the answers to these questions in the PST. The remaining questions (a past exam question) will be solved by your PST group in the class under the guidance of the tutor.

Wavefunctions, Dirac notation and Change of Basis

In Chapter 1 of the course, we introduced Dirac notation and column vector representations for states. We also introduced matrix representations for operators, showed how we can change the basis of our representations and finally saw how wave-functions and Dirac kets are related.

The **aim** of this PST is for you to gain confidence in working with Dirac notation, wave-functions and column-vector representations and the relationship between them, and apply the methods of basis change to a concrete example.

1. **[Attempt before the PST]** Consider a one-dimensional particle of mass m in a potential well, which is zero between $x = 0$ and $x = L$ and infinite elsewhere. This is the well-known “particle in a box” and the system’s orthonormal energy eigenfunctions are:

For $n = 1, 2, 3, \dots$

$$\phi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi}{L}x\right)$$

inside the well and $\phi_n(x) = 0$ elsewhere.

Consider the state $|\psi_1\rangle$ which has the following wavefunction:

$$\psi_1(x) = \frac{1}{\sqrt{14}} (\phi_1(x) + 3\phi_2(x) + 2\phi_3(x))$$

Represent $|\psi_1\rangle$ as a 3-element column vector. What assumption are we making when we express the state in this form?

Solution: We are going to do this step by step and first introduce Dirac notation for the energy eigenstates. Let us use $|\phi_n\rangle$ as the Dirac vector corresponding to wave-function $\phi_n(x)$. Recall that the relationship between $\phi_n(x)$ and $|\phi_n\rangle$ is

$$\langle x|\phi_n\rangle = \phi_n(x)$$

From that definition, if we have a superposition:

$$\sum_j c_j |\phi_j\rangle$$

in Dirac notation, the wave-function is:

$$\langle x|\sum_j c_j |\phi_j\rangle = \sum_j c_j \langle x||\phi_j\rangle = \sum_j c_j \phi_j(x)$$

In other words, when we have a superposition of wave-functions we can write the equivalent superposition, with the same coefficients, in Dirac notation. Hence

$$|\psi_1\rangle = \frac{1}{\sqrt{14}} (|\phi_1\rangle + 3|\phi_2\rangle + 2|\phi_3\rangle)$$

We would now like to express $|\psi_1\rangle$ as a column vector, using $|\phi_n\rangle$ as our orthonormal basis, but be careful, the index n runs from 1 to infinity! We cannot write down an infinite-sized vector, so we **truncate**, we will **assume** that we will only consider states whose coefficients c_n for all higher eigenstates than $n = 3$ are zero.

E.g. only states of the form:

$$c_1|\phi_1\rangle + c_2|\phi_2\rangle + c_3|\phi_3\rangle.$$

Then we can use a 3-dimensional column vector representation, via the standard representation:

$$|\phi_1\rangle \rightarrow \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad |\phi_2\rangle \rightarrow \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad |\phi_3\rangle \rightarrow \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Hence the column vector representation for $\psi_1(x)$ is

$$\sqrt{\frac{1}{14}} \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$$

We will use this **truncation** trick later in the course to derive matrix representations for quantum harmonic oscillator operators, which are infinitely dimensional.

2. **[Attempt before the PST]** Now consider the state $|\psi_2\rangle$ which has the following wave-function:

$$\psi_2(x) = \frac{1}{\sqrt{2}} (\phi_1(x) + i\phi_3(x))$$

Represent $|\psi_2\rangle$ as a 3-element column vector, and calculate the inner product between $\psi_1(x)$ and $\psi_2(x)$.

Solution: This is deliberately phrased to make the student think about the relationship between inner products in the three representations. We saw in Chapter 1 that they are equal, regardless of representation i.e. the inner product between wave-functions $\psi_1(x)$ and $\psi_2(x)$ is equal to the inner product between $|\psi_1\rangle$ and $|\psi_2\rangle$ and also their respective column vector representations.

We can simply write down the column vector representation of $|\psi_2\rangle$ as:

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ i \end{bmatrix}$$

The column vector representation gives us a convenient way to compute the inner product between $\psi_1(x)$ and $\psi_2(x)$ without solving any integrals.

$$\langle \psi_1 | \psi_2 \rangle = \frac{1}{\sqrt{14}} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ i \end{bmatrix} = \frac{1 + 2i}{\sqrt{28}}$$

3. **[Attempt before the PST]** To explore change of basis, we are going to consider an invented physical property. Imagine that you have discovered a new physical property of particles in wells. You call it the Lowity, represented with the greek Lambda Λ , because you realise that Lowity is zero for all energy eigenstates of the well except for the lowest states $n = 1, 2, 3$. For this reason we can represent Lowity with a 3x3 matrix, making the same truncation of the vector space that we did in questions 1 and 2.

You determine that the matrix representation of Lowity with respect to the states $|\phi_1\rangle$, $|\phi_2\rangle$, $|\phi_3\rangle$ is:

$$\Lambda = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 6 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

You are told that this matrix has the following set of eigenvalues: 2, 4, 6. Calculate the associated eigenvectors of the matrix Λ , labelled $|\chi_1\rangle$, $|\chi_2\rangle$ and $|\chi_3\rangle$ for eigenvalues 2, 4 and 6 respectively.

Solution:

This is an exercise in computing normalised eigenvectors.

Summary of results:

We are going to choose to label our eigenvectors as follows:

- Eigenvalue 2: Eigenvector $|\chi_1\rangle$
- Eigenvalue 4: Eigenvector $|\chi_2\rangle$
- Eigenvalue 6: Eigenvector $|\chi_3\rangle$

Substituting these into the eigenvalue equation, solving the resultant simultaneous equations and normalising we get the following eigenvectors and their corresponding column vectors:

$$|\chi_1\rangle = \frac{1}{\sqrt{2}}(|\phi_1\rangle - |\phi_3\rangle) \rightarrow \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$|\chi_2\rangle = \frac{1}{\sqrt{2}}(|\phi_1\rangle + |\phi_3\rangle) \rightarrow \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$|\chi_3\rangle = |\phi_2\rangle \rightarrow \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

NB Since eigenvectors are defined only up to a global phase, students might calculate different vectors here, in particular $|\chi_1\rangle$ with the minus sign in the other place. This is not incorrect, but it leads to different matrices below.

Calculation method:

When we know the eigenvalue, we can compute the eigenvector by solving the eigenvalue equation:

$$\begin{bmatrix} 3 & 0 & 1 \\ 0 & 6 & 0 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 2 \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

Hence

$$3a + c = 2a$$

and

$$6b = 2b$$

Hence $b = 0$ and $a = -c$.

We then have an eigenvector of the form:

$$a \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

a can be found by normalising using the standard method.

$$\langle \chi_1 | \chi \rangle = 1 = |a|^2 \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = 2|a|^2$$

We therefore need to solve the equation:

$$2|a|^2 = 1$$

or

$$|a|^2 = \frac{1}{2}$$

Any complex solution to this equation would be satisfactory, due to the **global phase freedom** of quantum states. This tells us that given any wave-vector $|\psi\rangle$, if we multiply it by a global phase $e^{i\theta}$ for any value θ the state $e^{i\theta}|\psi\rangle$ has identical physical properties to $|\psi\rangle$. Therefore we could choose any solution:

$$a = \frac{e^{i\theta}}{\sqrt{2}}$$

but conventionally, we set a to be real (and θ to be zero).

Hence, the normalised eigenvector in column vector form is:

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

In Dirac notation, the state is

$$|\chi_1\rangle = \frac{1}{\sqrt{2}}(|\phi_1\rangle - |\phi_3\rangle)$$

We follow the same approach for the other two eigenvectors:

$$\begin{bmatrix} 3 & 0 & 1 \\ 0 & 6 & 0 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 4 \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

and obtain simultaneous equations:

$$3a + c = 4a$$

$$6b = 4b$$

Hence $b = 0$ as above and

$$c = a$$

and the normalised eigenvector is:

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

So in Dirac notation, $|\chi_2\rangle = \frac{1}{\sqrt{2}}(|\phi_1\rangle + |\phi_3\rangle)$

Finally we consider the case where the eigenvalue is 6

$$\begin{bmatrix} 3 & 0 & 1 \\ 0 & 6 & 0 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 6 \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

Now we have:

$$3a + c = 6a$$

$$6b = 6b$$

and this time we need to consider the third equation too:

$$a + 3c = 6c$$

From first equation:

$$c = 3a$$

From the third:

$$a = 3c$$

These can only be simultaneously satisfied if $a = c = 0$.

The second equation tells us nothing about b , but we can fix b via normalising.

Hence our final eigenvector χ_3 has the form:

$$\begin{bmatrix} 0 \\ b \\ 0 \end{bmatrix}$$

which when normalised gives us:

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$|\chi_3\rangle = |\phi_2\rangle$$

4. **[Attempt before the PST]** To study the properties of Lowity, we would like to express states $|\psi_1\rangle$ and $|\psi_2\rangle$ with respect to the basis $|\chi_1\rangle$, $|\chi_2\rangle$ and $|\chi_3\rangle$. Construct a basis change matrix S that transforms from a representation in the $|\phi_1\rangle$, $|\phi_2\rangle$, $|\phi_3\rangle$ basis to the $|\chi_1\rangle$, $|\chi_2\rangle$, $|\chi_3\rangle$ basis.

Solution: We recall from the lectures that the basis change matrix S has matrix elements

$$S_{j,k} = \langle \chi_j | \phi_k \rangle$$

where j is the row label and k is the column label.

So to construct S we have 9 inner products to calculate. It is easiest to do this using the column vector representations we have already derived:

$$\langle \chi_1 | \phi_1 \rangle = \frac{1}{\sqrt{2}} [1 \ 0 \ -1] \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}}$$

$$\langle \chi_1 | \phi_2 \rangle = \frac{1}{\sqrt{2}} [1 \ 0 \ -1] \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = 0$$

$$\langle \chi_1 | \phi_3 \rangle = \frac{1}{\sqrt{2}} [1 \ 0 \ -1] \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = -\frac{1}{\sqrt{2}}$$

$$\langle \chi_2 | \phi_1 \rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}}$$

$$\langle \chi_2 | \phi_2 \rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = 0$$

$$\langle \chi_2 | \phi_3 \rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}}$$

$$\langle \chi_3 | \phi_1 \rangle = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = 0$$

$$\langle \chi_3 | \phi_2 \rangle = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = 1$$

$$\langle \chi_3 | \phi_3 \rangle = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = 0$$

Hence

$$S = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & 1 \\ 0 & \sqrt{2} & 0 \end{bmatrix}$$

Note that the rows of S have the forms of the column vector representations of $|\chi_n\rangle$. This is not an accident, as each row has the form $\langle \chi_j | \phi_1 \rangle \langle \chi_j | \phi_2 \rangle \langle \chi_j | \phi_3 \rangle \dots$ and the elements of the column vector representation of $|\chi_k\rangle$ in the $|\phi_j\rangle$ basis are $\langle \phi_j | \chi_k \rangle$. More generally S could be formed by stacking the transverse conjugate vector representations of $|\chi_k\rangle$ (our example here had no complex elements in the new basis).

5. **[Attempt before the PST]** Transform $|\psi_1\rangle$ (which you wrote down in the $|\phi_1\rangle, |\phi_1\rangle, |\phi_1\rangle$ basis in your answer to question 1) into the $|\chi_1\rangle, |\chi_2\rangle, |\chi_3\rangle$ basis using the S matrix. Express your final answer in Dirac notation.

Solution: We apply the matrix S to the column vector we derived in question 1:

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & 1 \\ 0 & \sqrt{2} & 0 \end{bmatrix} \sqrt{\frac{1}{14}} \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} = \sqrt{\frac{1}{28}} \begin{bmatrix} -1 \\ 3 \\ 3\sqrt{2} \end{bmatrix}$$

which in Dirac notation is:

$$|\psi_1\rangle = \sqrt{\frac{1}{28}} \left(-|\chi_1\rangle + 3|\chi_2\rangle + 3\sqrt{2}|\chi_3\rangle \right).$$

6. [Past exam question 2023-24]:

- (a) The sets of states $\{|\chi_m\rangle\}$ and $\{|\psi_n\rangle\}$ are two independent orthonormal sets of basis states such that a quantum state $|\Psi\rangle$ can be written $|\Psi\rangle = \sum_n c_n |\psi_n\rangle$ and $|\Psi\rangle = \sum_m d_m |\chi_m\rangle$. Use the closure relation to show that the change in representation between column vectors $\mathbf{c}$ and $\mathbf{d}$ can be expressed as $\mathbf{d} = S\mathbf{c}$ where S is a matrix with elements $S_{mn} = \langle\chi_m|\psi_n\rangle$.

Solution: This question is essentially asking you to reproduce the derivation given in lectures. It gives you a big hint to use the closure relation, and the closure relation (so don't worry if you can't remember it) is given in the materials booklet for the exam.

It is

$$\mathbb{1} = \sum_n |\phi_n\rangle\langle\phi_n|.$$

NB In lectures we used $\hat{I}$ to represent the identity operator, but the blackboard $\mathbb{1}$ is also commonly used here and in the notes.

Let us start with $|\Psi\rangle$ written in the $|\psi_n\rangle$ basis, and insert the closure relation in the $|\chi_m\rangle$ basis

$$\begin{aligned} |\Psi\rangle &= \sum_n c_n |\psi_n\rangle = \sum_n c_n \mathbb{1} |\psi_n\rangle \\ &= \sum_n c_n \sum_m |\chi_m\rangle\langle\chi_m| |\psi_n\rangle \\ &= \sum_{n,m} c_n |\chi_m\rangle\langle\chi_m|\psi_n\rangle = \sum_{n,m} \langle\chi_m|\psi_n\rangle c_n |\chi_m\rangle \end{aligned}$$

This must equal:

$$|\Psi\rangle = \sum_m d_m |\chi_m\rangle$$

hence

$$d_m = \sum_n \langle \chi_m | \psi_n \rangle c_n$$

This is a matrix multiplication written elementwise

$$\mathbf{d} = S\mathbf{c}$$

where $S_{m,n} = \langle \chi_m | \psi_n \rangle$

- (b) Show that the matrix representation A_χ of an operator $\hat{A}$ in the $\{|\chi_m\rangle\}$ basis is related to A_ψ , its representation in the $\{|\psi_n\rangle\}$ basis, by

$$A^{(\chi)} = S A^{(\psi)} S^\dagger.$$

Solution: Now we are going further than the live lectures, and being asked to derive the transformation rule for operators, which was covered in one of the videos and also in the notes.

It is a similar principle to the previous derivation, except now we need to use the closure relation *twice* inserting it on the left and on the right. Although bookwork, it is an intricate proof, and was a challenging part of that year's exam.

Our starting point is the expansion of $\hat{A}$ in the $|\psi_n\rangle$ basis:

$$\hat{A} = \sum_{j,k} A_{j,k}^{(\psi)} |\psi_j\rangle \langle \psi_k|$$

Now insert $\mathbb{1}$ either side of $\hat{A}$:

$$\hat{A} = \mathbb{1} \hat{A} \mathbb{1} = \sum_{j,k} A_{j,k}^{(\psi)} \mathbb{1} |\psi_j\rangle \langle \psi_k| \mathbb{1}$$

Now we expand the $\mathbb{1}$ using the closure relation taking care not to reuse any dummy indices for the sums (a common mistake)

$$\begin{aligned} &= \sum_{j,k} A_{j,k}^{(\psi)} \sum_r |\chi_r\rangle \langle \chi_r| |\psi_j\rangle \langle \psi_k| \sum_s |\chi_s\rangle \langle \chi_s| \\ &= \sum_{j,k,r,s} A_{j,k}^{(\psi)} \langle \chi_r | \psi_j \rangle \langle \psi_k | \chi_s \rangle |\chi_r\rangle \langle \chi_s| \end{aligned}$$

We are free change the order of the scalars and we reorder them so that the terms are in the correct order to represent elementwise matrix multiplication:

$$= \sum_{j,k,r,s} \langle \chi_r | \psi_j \rangle A_{j,k}^{(\psi)} \langle \psi_k | \chi_s \rangle | \chi_r \rangle \langle \chi_s |$$

Now we compare with the definition of $A^{(\chi)}$

$$\hat{A} = \sum_{r,s} A_{r,s}^{(\chi)} | \chi_r \rangle \langle \chi_s |$$

Hence

Now comparing our two expressions, we have:

$$A_{r,s}^{(\chi)} = \sum_{j,k} \langle \chi_r | \psi_j \rangle A_{j,k}^{(\psi)} \langle \psi_k | \chi_s \rangle$$

and thus

$$A^{(\chi)} = S A^{(\psi)} S^\dagger.$$

where $S_{r,j} = \langle \chi_r | \psi_j \rangle$ and (consistently) $(S^\dagger)_{k,s} = \langle \psi_k | \chi_s \rangle$.

To go from S to $S^\dagger$ you need to complex conjugate (swap the bra and ket) and then transpose (also swap the labels).